

COMP2200/COMP6200 Practical Exercise (Week 6): Decision Trees, CN2, & Random Forests — Hands-on & Orange

School of Computing

Abstract

You'll first learn how decision trees split on numeric attributes by *physically* partitioning a table of cards with paddle-pop sticks (simulating impurity minimisation). Next, you will experience rule learning with a simplified CN2 “beam search” game. Finally, you'll reproduce the same ideas in Orange (visual data mining) with proper evaluation and stacking. The tiny dataset `watchband.csv` (18 instances) keeps arithmetic simple while still exposing all the ideas.

What you need (per group)



- Printed cards: one per row of `watchband.csv`
- Lots of stirring sticks, which will be used to divide the cards
- A lot of space on a table

You have 18 cards that represent different people who have come into your watchband store. You have superpowers that let you detect their age and their income perfectly. You would like to predict what colour watchband people buy based on your transaction history.

Decision trees with paddle-pop sticks

Setup

Lay the cards on the table in a grid with their Income as the x direction and Age as the y direction. Note that you don't need to have the \$100,000 income be twice as far along as the \$50,000 income. Just make sure that when you move along the x direction, that the Income increases, and when you move along the y direction that Age increases.

You have some stirring sticks. You are going to place the stirring sticks either in the x direction or the y direction.

Top-level split for a decision tree

Looking at the layout, where do you think would be a good place to put a stirring stick so that you split the dataset into two groups so that the groups are more "pure"? i.e. Where could you place the stirring stick so that each group is closer to the ideal of "each section just has one colour"?

Come up with a few good options. Now score them.

How to score a split

- If you placed your stick vertically, start with the cards on the left. If you placed your stick horizontally, start with the cards above the stick
- Count up the number of orange cards (r), count up the number of blue cards (b), count up the number of green cards (g). Add up the total number of cards: $N = r + b + g$
- The Gini index for a set is $G = 1 - \frac{r^2 + b^2 + g^2}{N^2}$
- Repeat with the other half.

Decide which split was the best (based on the scoring criteria). When you have chosen something, insert a line of coffee stirrers to split the data set in half.

Build a depth-2 tree

If you have two people on your team, have one person look at one side of the dataset and another person look at the other. Just looking at your side, work out where to make a good split: use the same procedure as before.

(If you only have one person on your team, do one side and then the other.)

Stop early if a node is pure or tiny (e.g. $n \leq 2$).

Build a depth-3 tree

Repeat the process on the four regions you now have. You might or might not end up with eight regions, because you could end up with pure regions, or tiny regions.

When you have finished, take a photograph and draw your divisions as a decision tree. Record the class distribution at each leaf.

CN2 rule induction

- Use **WRAcc** when you care about subgroups (lift x coverage).
- Use **Laplace** when you care about classification accuracy with small samples.
- Use **Entropy** to mirror tree-style purity.

In Orange, pick the measure in CN2's "Evaluation measure" setting.
We'll use WRAcc in this prac.

CN2 learns *if-then* rules via separate-and-conquer. We'll approximate it with a small beam search and a simple rule quality measure.

Key ideas

- **Hypotheses** are conjunctions of simple conditions ("literals"): e.g. **Age > 24 & Income ≤ 90 000 ⇒ Green**.
- **Star operator**: specialise a rule by adding one more literal (tighter rectangle on the grid).
- **Beam width B** : keep only the top B candidates each round.
- **Covering**: once a high-quality rule for target class c is found, *remove* the covered c -cards and repeat to get a ruleset.

Scoring rules

We'll use **Weighted Relative Accuracy (WRAcc)** to reward both purity and coverage for target class c :

$$\text{WRAcc} = \frac{n}{N} \left(\frac{p}{n} - \frac{P}{N} \right) = \frac{p}{N} - \frac{n}{N} \cdot \frac{P}{N},$$

where N is dataset size, P is the # of class- c examples in the whole dataset, n is the # covered by the rule and p is the # of covered examples of class c .

Worked examples

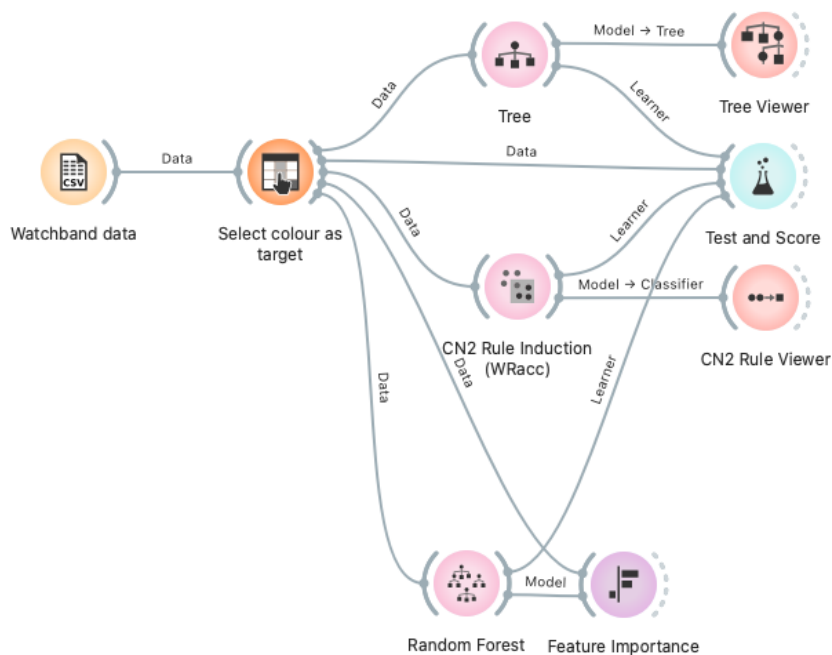
- For the rule **Age > 24 & Income ≤ 90 000 ⇒ Green**: $(p, n, P, N) = (5, 6, 6, 18)$ so $\text{WRAcc} = \frac{6}{18} \left(\frac{5}{6} - \frac{6}{18} \right) = \frac{1}{6} \approx 0.167$.
- For the rule **16 < Age ≤ 24 ⇒ Blue**: $(p, n, P, N) = (5, 6, 6, 18)$ so $\text{WRAcc} \approx 0.167$.
- For the rule **Age ≤ 16 ⇒ Orange**: $(p, n, P, N) = (3, 3, 6, 18)$ so $\text{WRAcc} = \frac{1}{9} \approx 0.111$.

Procedure (per target class)

1. Choose a **target class** (e.g. start with **Blue**). Record P and N once.
2. **Initial beam**: all 1-literal rules using the possible splits you could have done at the start of the decision trees section. Keep the top $B=3$ by WRAcc.
3. **Specialise**: for each kept rule, add one more literal (either on Age or Income) to form new candidates that still cover at least one target card. Re-score; keep top B .
4. **Stop** after two rounds without WRAcc improvement or once accuracy $p/n \geq 0.8$ with $n \geq 3$.
5. Add the best rule to your ruleset for the target, *remove* its covered target-class cards from the table, and go back to step 1 for the remaining positives of that class. Then switch to the next class.

Rebuild it in Orange

Let's also add a random forest so that we can have a very accurate model. We can't get a full explanation from it, but we can see which features are important in explaining the outcome.



You will need to set the Evaluation measure on the CN2 Rule Induction widget to WRAcc.

- Look at the Tree Viewer. Does the tree match the tree you drew?
- Look at the CN2 Rule Viewer. Do the rules match what you came up with?
- Look at the feature importances. What is the most important feature?

Quiz questions

Carry on with the quiz questions for this week.

A Data table for watchband.csv

#	Age	Income	Colour
1	12	0	Orange
2	14	10	Orange
3	15	0	Orange
4	17	500	Blue
5	18	10 000	Orange
6	19	30 000	Blue
7	21	0	Blue
8	21	20 000	Blue
9	23	30 000	Blue
10	25	40 000	Green
11	27	100 000	Orange
12	30	80 000	Green
13	40	150 000	Orange
14	41	60 000	Green
15	60	100 000	Green
16	62	40 000	Blue
17	63	60 000	Green
18	65	30 000	Green